

MANDEEP KHATRI

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EDUCATION

Alabama A&M University

Expected Graduation: May 2028

Bachelor of Science in Computer Science GPA: 3.80

Huntsville, AL

- **Relevant Coursework:** Data Structures, Java, C++, Python, Intro to Digital Logic Design, Calculus I & II
- **Honors:** AAMU Full Merit Scholarship, EICOP 2025 Finalist

SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, SQL (PostgreSQL, MySQL), HTML, CSS

Frameworks & Tools: Flask, React.js, Streamlit, LangChain, FAISS, PyMuPDF, Docker, Git, Firebase, SQLite, Notification API

AI/ML & Cloud: AWS, GCP (Vertex AI, Gemini API), TensorFlow, PyTorch, NLP, Deep Learning

CS Fundamentals: OOP, Data Structures & Algorithms, Complexity Analysis, RAG, Prompt Engineering

CERTIFICATIONS

Google Cloud Certified – Associate Cloud Engineer (Career Jumpstart Summer 2025)

WORK EXPERIENCE

Academic Teaching Assistant

January 2025 – Present

Alabama A&M University

Huntsville, AL

- Instructing **50+** students in **C++, Python and Calculus I**, driving a **15%** improvement in performance.
- Designing and delivering interactive lessons, coding labs, and practice materials to strengthen problem-solving skills.
- Leading live programming demos and project-based workshops, boosting engagement and retention.

AAMUTeach Program

January 2025 – May 2025

Alabama A&M University

Huntsville, AL

- Taught **STEM** curriculum to elementary and middle school students, introducing programming, problem solving, and teamwork through hands-on projects.
- Developed engaging lesson plans that simplified technical concepts, improving student understanding while building leadership and communication skills.

Cloud Computing Intern

January 2023 – March 2023

Verisk

Kathmandu, Nepal

- Developed and implemented a cloud-based IoT monitoring system using **Python** and **C++**, enhancing real-time data collection and improving data processing efficiency by **30%**.
- Created a responsive web-based dashboard with **HTML, CSS, JavaScript**, improving user interaction & usability by **20%**.
- Optimized system performance through testing and debugging, reducing latency by **25%** & enhancing system reliability.

PROJECTS

Smart To-Do List | Flask, SQLite, JavaScript, HTML/CSS, Notification API

August 2024

- Built a task manager using **Flask & SQLite**, supporting CRUD, priorities, & deadlines for **100+** tasks with persistent storage.
- Developed UI with **Vanilla JS, HTML/CSS**, and **Notification API**, enabling real-time updates and deadline alerts.

LLM-Powered Q&A Chatbot | Python, Gemini API, LangChain, FAISS, Streamlit

February 2024

- Built a Q&A chatbot using **Python, Gemini 1.5 Pro API (via Vertex AI)**, and **LangChain**, processing **100+** pages of course PDFs with **PyMuPDF**, text splitting, and embedding via **Gemini**; reached **85%** manual QA accuracy.
- Implemented **RAG** pipeline with **FAISS** (dense retrieval) and **Streamlit** UI, optimizing prompt templates and reducing avg. response latency by **40%** vs. keyword search baseline.

ACTIVITIES

Google Cloud Career Jumpstart HBCU Fellow

May 2025 – August 2025

HBCU Possibilities Summit

January 2025 – May 2025

Apprenticeship Universe Program — HBCU 20x20

May 2025

TracerFire Cybersecurity Competition – Sandia National Laboratories

September 2024